Ask Small VLMs Only What They Can See: Semantic-to-Metric Navigation with Local Models

Anonymous ICCV submission

Paper ID *****

Abstract

Small open vision-language models (VLMs) are poor embodied navigators when they must plan in text. Prompted with the EB-Navigation benchmark's own planning interface, Qwen3-VL-4B solves 55.0% of the test episodes and 25.0% when the target starts behind the agent. The cause is neither recognition nor motion planning: a geometric planner given the target position succeeds in 91.3% of the same episodes, and the model boxes a visible target of a named or implied category in 69.3–78.7% of single views. The failure lies in the interface, which makes the model track the target, the room, and its own progress through generated text. We propose S2M (Semantic-to-Metric), a training-free agent that asks the VLM only what the current image can answer, namely whether the target is visible and where it is in the frame. A metric hypothesis memory lifts these answers with depth and pose, accumulates their evidence over distinct viewpoints, discounts hypotheses that should be visible but are not detected, and selects targets with an optimal-search index; an occupancy-map planner turns the selection into actions. With the model at 4-bit on one 8 GB GPU, S2M reaches 79.4% (+24.4 pp over the benchmark's agent), 82.1% on Long-horizon episodes, and 76.5% without a depth sensor or simulator pose, using monocular depth whose scale is fixed by the floor in every frame. With Qwen2.5-VL-3B it reaches 70.0% against 20.3%. On the 1800-episode RoboTHOR validation split, which requires an explicit stop decision, S2M reaches 59.6% success, above ten published zero-shot agents.

1. Introduction

Multimodal LLM agents now follow open-ended navigation instructions in simulated homes, but the strongest results come from frontier models behind an API. On EB-Navigation [68], GPT-4o succeeds in 57.7% of the episodes, while the open models with at most 8B parameters reach 14.0–29.3% overall and 0–5% on the Long-horizon subset, in which the target starts behind the agent. Robots need such small models, which run on board or on a consumer GPU; the training-free gains reported on this benchmark use 8B and larger models with memory, harnesses, or tool libraries [30, 42, 78].

We first ask what breaks when a small VLM drives the agent. Motion planning does not: a geometric planner given the target position solves 91.3% of the test episodes. Recognition is imperfect but usable: from a single view, Qwen3-VL-4B boxes a visible target whose category the instruction names or implies in 69.3–78.7% of cases. The failure sits in between. The benchmark's agent asks the VLM for a multi-step JSON plan at every replanning step, so the model must track where the target was, what the room looks like, and which way it faces, all through generated text (Fig. 1a). Prompted agents lose targets they have already seen, collide with furniture, and in Long-horizon episodes sometimes never turn toward the target (Sec. 3).

These findings suggest a different division of labor. A small VLM should answer only questions that the current image can settle, and the spatial state of the agent should live outside the model. S2M (Semantic-to-Metric, Fig. 1b) asks the VLM two things per frame: is the target visible, and where is it. Each answer is lifted into a metric target hypothesis with depth and pose. Hypotheses gain evidence from distinct viewpoints and lose it when they should be visible but are not detected; a search index picks which one to visit, and an occupancy-map planner computes the motion.

Changing the interface changes the outcome (Fig. 2). With Qwen3-VL-4B at 4-bit, S2M succeeds in 79.4% of the EB-Navigation test episodes, +24.4 pp over the benchmark's agent with the same model, and in 82.1% of the Long-horizon episodes, 1.3 pp below the oracle. With Qwen2.5-VL-3B [3] the gain is +49.7 pp. S2M needs 12% fewer steps on the episodes that both agents solve. On the RoboTHOR ObjectNav validation split [12], where the agent must decide when to stop, S2M reaches 59.6% success.

From RGB alone, with monocular depth scaled in every frame by the floor and pose from the executed actions, S2M reaches 76.5%, +21.5 pp over the benchmark's agent

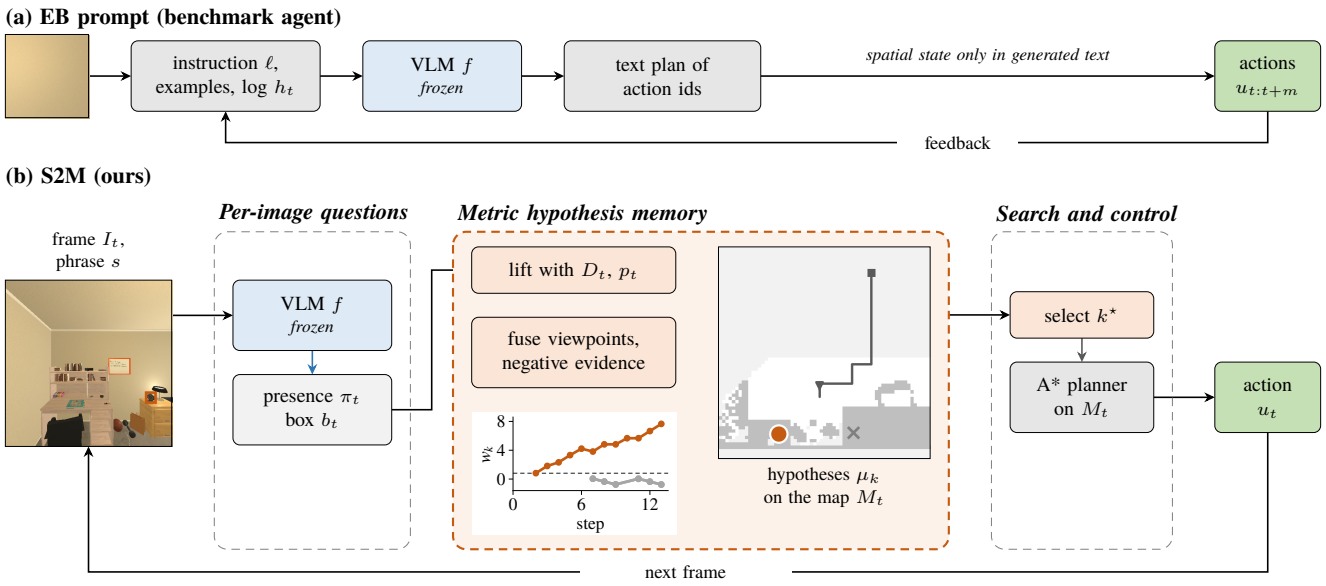


Figure 1. S2M moves the agent's spatial state out of generated text into an explicit metric memory. **(a)** The EB prompt maps I_t, the instruction ℓ, and the action log h_t to a text plan $u_{t:t+m}$ with the VLM f. **(b)** S2M asks f only for the presence π_t and the box b_t of the phrase s, lifts boxes with depth D_t and pose p_t into hypotheses μ_k with evidence w_k (Eqs. (5), (7) and (8)), selects $k^\star$ (Eq. (9)), and plans the action u_t with A* on the map M_t. Blue: frozen VLM; orange: S2M; green: actions; grey: inputs and inherited parts. Thumbnails from a Long-horizon test episode: the target (orange) passes the commit threshold (dashed), and two look-alikes (grey) fall below zero.

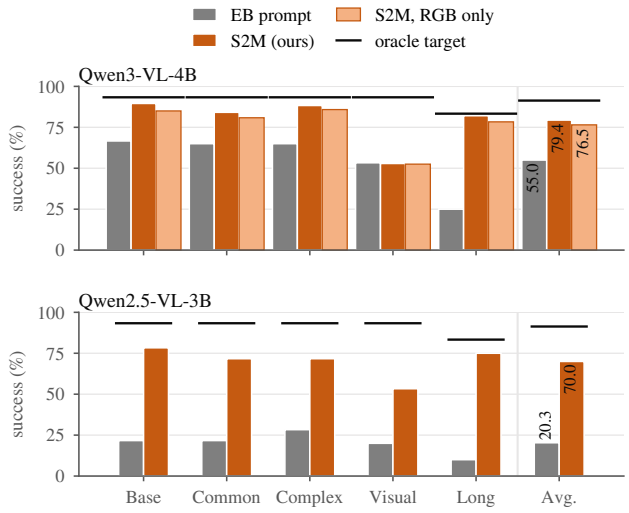


Figure 2. EB-Navigation test success with the same local model. The benchmark's agent (grey) against S2M (orange) and S2M without a depth sensor or simulator pose (light orange). Ticks: geometric planner given the true target position.

(Sec. 5.3).

Contributions. (1) A controlled diagnosis of small-VLM navigators on EB-Navigation, showing that they fail at carrying spatial state through text rather than at recognition or motion planning. (2) S2M, a training-free semantic-to-metric interface: per-image presence and box queries, metric target hypotheses fused across viewpoints with positive and negative evidence, an optimal-search selection rule, and a per-frame ground-plane scale that lets it run from RGB alone. (3) Full-split evaluations on EB-Navigation and RoboTHOR with two model generations and paired statistics.

2. Related Work

MLLM agents for embodied tasks. EmbodiedBench [68] evaluates multimodal LLMs on high- and low-level embodied tasks through a standardized action interface [33, 47]. Agents are improved by embodied pre-training, online reinforcement learning, or policy-level reasoning [6, 39, 56, 75], or at test time with reflection, experience, memory, or tools [7, 30, 32, 42, 52, 53, 60, 76, 78]. On EB-Navigation, the strongest agents let an 8B or larger model call perception and mapping tools [78] or render the target into a bird's-eye view [59], and keep the model in charge of the plan. S2M removes planning from the model: the VLM is a sensor whose answers feed an explicit estimator and planner.

Zero-shot object navigation and semantic memory. Modular zero-shot ObjectNav combines open-vocabulary perception, a map of explored space, and a planner. CoW [15] pairs open-vocabulary localization with frontier exploration; later systems build vision-language value maps [72], scene graphs reasoned over by an LLM [20,

70], goal graphs [71], world models [46], cognitive state machines [5], or structured representations queried by a VLM [79]; instruction-following agents ground language in maps and scene graphs [41, 49, 62]. Open-vocabulary 3D scene graphs [18, 63], online semantic maps [22], language-embedded fields [25, 31, 48], and embodied memories [1, 16, 69] attach semantics to geometry. Closest to our memory, ApexNav fuses target-centric detections over time [74], and AECNav treats expected but absent detections as negative evidence [38]. S2M sits at the minimal end of this spectrum: a small generative VLM is its only perception model, and its memory is a short list of metric target hypotheses rather than a dense semantic map or scene graph.

Verification and search. Set-of-Mark prompting [66] and PIVOT [45] make VLM decisions spatially explicit, navigation can be posed as visual question answering over depth-derived options [17], and verification, test-time computation, and calibrated uncertainty improve embodied decisions [9, 11, 28, 36, 50, 51]. S2M spends extra queries, a magnified crop or a tiled view, only where the evidence is weak, and chooses among candidate targets with the probability-to-cost index of optimal sequential search [24].

Geometry from images. Learned reconstruction and depth priors recover geometry from RGB [37, 43, 44, 55, 61, 67], and the choice of sensors can change the conclusions of embodied evaluations [4, 10, 58]. Monocular visual odometry fixes its scale from the ground plane and the known camera height [54]; S2M applies the same constraint to every frame of a monocular metric depth network. Concurrently, Li *et al*. [34] run Qwen3-VL-4B zero-shot and accumulate its answers as per-direction scores in a bird's-eye field for vision-language navigation; S2M lifts boxes to metric target hypotheses for object navigation.

3. Problem and Diagnosis

Task. An episode gives an instruction ℓ that identifies one object with floor position $x^\star \in \mathbb{R}^2$. At step t the agent observes an RGB frame I_t, depth D_t, a pose $p_t = (x_t, \theta_t, \phi_t)$ of floor position, heading, and camera tilt, and the success flag o_{t-1} of its last action, and it chooses $u_t \in \mathcal{U}$: four 0.25 m translations, two 90° turns, and two 30° tilts. The episode is successful if

$$\exists\, t \le T : \ \|x_t - x^\star\| \le 1\,\mathrm{m}, \qquad T = 20. \tag{1}$$

EB-Navigation [68] poses this task in single AI2-THOR rooms [26]; its test set has five subsets of 60 episodes. *Base* names the category, *Common* describes the object by its use, *Complex* embeds it in a longer instruction, *Visual* describes only its appearance, and *Long* repeats Base with the agent turned away from the target.

The benchmark's agent. The benchmark's own agent, the *EB prompt*, maps the image and its text history $h_t = (u_{0:t-1}, o_{0:t-1})$ directly to a plan,

$$(u_t, \ldots, u_{t+m}) = f(I_t, \ell, h_t), \tag{2}$$

which is executed until an action fails. All spatial state, where the target was, what the room contains, and how the agent has turned, must be carried by h_t and the generated reasoning. S2M factorizes the same decision:

$$\begin{gathered}(\pi_t, b_t) = f(I_t, s), \\ \mathcal{B}_t = g(\mathcal{B}_{t-1}, \pi_t, b_t, D_t, p_t), \qquad u_t = \rho(\mathcal{B}_t, p_t),\end{gathered} \tag{3}$$

where s is a target phrase, f answers only questions about I_t, and the metric belief $\mathcal{B}_t$ and the planner ρ hold all state and every decision (Sec. 4). The configuration of S2M was fixed on a development split of 150 episodes that we built with the benchmark's generator on 30 iTHOR rooms not used by the test set.

Planning is not the bottleneck. Replacing the target belief of S2M by the true $x^\star$, with the same occupancy map and planner, solves 91.3% of the 300 test episodes (93.3% on each of the first four subsets and 83.3% on Long).

Single views are informative but unreliable. For each test episode of the first four subsets we render the start view and the three other 90° headings and query the VLM with the phrase s. Qwen3-VL-4B places a box with IoU$\geq$0.3 on the visible target in 78.7, 69.3, 77.3, and 49.3% of the views on Base, Common, Complex, and Visual (75 visible instances each). One answer is useful evidence, but not something to act on alone.

The prompted agent loses spatial state. We label every failed episode by the first matching rule: the target never enters the view; final distance at most 1.5 m (*near miss*); the target is out of view in the last five steps (*lost after seen*); at least 30% of the actions are rejected (*blocked*); at least three immediate reversals; otherwise approach failure. Of the 135 episodes that the EB prompt fails with Qwen3-VL-4B, 43 lose a target that was in view, 26 are blocked by collisions, and 34 end within 1.5 m (Fig. 3b); in 12 Long episodes the agent never brings the target into view.

Design principles. (i) Ask the VLM only what the current image settles, with no transcript to extend. (ii) Keep the target, the map, and the search state in an explicit metric memory. (iii) Treat every answer as weak evidence and act on evidence aggregated over viewpoints.

4. Semantic-to-Metric Navigation

S2M instantiates the factorization of Eq. (3). The belief $\mathcal{B}_t = (\mathcal{H}_t, M_t)$ consists of a list of target hypotheses $\mathcal{H}_t$ and an occupancy map M_t; Fig. 1b shows one step.

4.1. Per-image questions

At the start of an episode the VLM f rewrites ℓ into a target phrase s: the category when the instruction names or implies

one, and the appearance words when it describes only how the object looks. At every step two queries about the current frame run in parallel. The presence probability is read from the first answer token,

$$\pi_t = \frac{P_f(\mathit{yes} \mid I_t, s)}{P_f(\mathit{yes} \mid I_t, s) + P_f(\mathit{no} \mid I_t, s)}, \tag{4}$$

and the second query returns a box b_t for s or none. Neither query contains the action history, the map, or earlier answers, so every answer can be checked against the image alone.

Lifting. A box with centre $c(b_t) = (u, v)$ is lifted with the median depth $\tilde{d}_t$ over its central half (half its width and half its height). With focal length φ and principal point (u_0, v_0) of the pinhole camera, tilt ϕ_t, and the forward and right unit vectors $e_f(\theta_t), e_r(\theta_t)$ of the heading,

$$q_t = x_t + \tilde{d}_t \Big[\big(\cos\phi_t - \tfrac{v-v_0}{\varphi} \sin\phi_t \big)\, e_f(\theta_t) + \tfrac{u-u_0}{\varphi}\, e_r(\theta_t) \Big], \tag{5}$$

which we write $q_t = \Pi_{p_t}^{-1}(c(b_t), \tilde{d}_t)$; lifts with $\tilde{d}_t > 6\,\mathrm{m}$ are discarded. The same back-projection, with the known camera height, labels each depth pixel as floor (height below 0.10 m) or obstacle (0.10–1.9 m) in a 5 cm occupancy grid M_t.

Observation weight. A lifted box enters the memory with weight

$$a_t = \begin{cases} \pi_t & \text{if the target is named and } \pi_t \geq \tfrac{1}{2}, \\ \pi_t^z & \text{otherwise, if } \pi_t^z \geq \tfrac{1}{2}, \\ 0.2\,\pi_t^z & \text{otherwise,} \end{cases} \tag{6}$$

where π_t^z is the presence probability of a *crop check*, a square crop of 2.5 times the box's longer side resized to 448 px. The crop is queried only when presence and box disagree or when the goal is appearance-only, so the extra cost is paid where the evidence is weak.

4.2. Metric hypothesis memory

Each hypothesis k stores a floor position μ_k, evidence w_k, and the poses V_k from which it was seen. An observation (q_t, a_t) is associated with the first hypothesis within $r = 0.5\,\mathrm{m}$. If there is one, then

$$\mu_k \leftarrow \frac{n_k \mu_k + q_t}{n_k + 1}, \qquad w_k \leftarrow w_k + \gamma_t\, a_t, \tag{7}$$

with n_k the number of fused observations and $\gamma_t = 0.25$ if some $v \in V_k$ has the heading of p_t and a position within 0.2 m of it, else $\gamma_t = 1$; otherwise a new hypothesis $(\mu, w) = (q_t, a_t)$ is created. Through γ_t, evidence grows with the number of distinct viewpoints rather than with the number of frames.

Negative evidence. A hypothesis that the current frame should show but that receives no detection loses evidence:

$$w_k \leftarrow w_k - \lambda \mathbb{1}\big[\mathrm{vis}_t(\mu_k)\big], \qquad \mathrm{vis}_t(\mu) \Leftrightarrow \bar{u} \in \Omega,\ \ d \leq 6\,\mathrm{m},\ \ D_t(\bar{u}) + 0.3 \geq d, \tag{8}$$

for every hypothesis without a detection within r, where $(\bar{u}, d) = \Pi_{p_t}(\mu)$ are the pixel and planar depth of μ in the current image, Ω is the image shrunk by a 20 px margin, and $\lambda = 0.4$; the last condition rules out occlusion by a nearer surface; hypotheses with $w_k \leq -1$ are removed. Absence outside the camera's view is never counted.

Commitment and exhaustion. The agent commits to a hypothesis once $w_k \geq \tau_c = 0.8$ and keeps it while $w_k \geq \tau_k = 0.4$; the hysteresis prevents one missed detection from abandoning a well-supported target. When the agent stands within 0.9 m of a committed hypothesis and the episode continues, which on EB-Navigation means that it is not the target, the hypothesis is marked exhausted. It stays in memory, so later detections of the same look-alike merge into it instead of creating a new target.

4.3. Search and control

Selection. Among the committed hypotheses $\mathcal{C}_t$ the agent visits

$$k^\star = \arg\max_{k \in \mathcal{C}_t} \frac{1 - e^{-w_k}}{c_k + 1}, \qquad c_k = \frac{\|\mu_k - x_t\|}{0.25\,\mathrm{m}}. \tag{9}$$

The numerator maps the evidence to the probability of at least one detection under a Poisson model with mean w_k, and the denominator counts the steps to reach and check μ_k. For locations with fixed inspection costs, inspecting in decreasing order of this probability-to-cost ratio is optimal [24]; S2M applies the rule greedily, recomputing the cost from the current position. It reduces to nearest-first for equal evidence and to strongest-first for equal cost. With nothing committed, the agent first completes a 360° look-around, then approaches the strongest hypothesis with $w_k \geq 0.35$, and otherwise moves to the observed free cell 0.75–3 m away that is farthest from all visited positions.

Planning. Actions come from A* over (position, heading) on M_t, whose edges are the benchmark's translations and turns. An edge e costs

$$C(e) = 1 + \frac{0.6}{4} \sum_{j=1}^{4} \mathbb{1}\big[M_t(e_j) \text{ unobserved}\big], \tag{10}$$

where $e_1, \ldots, e_4$ are points sampled along the edge; edges through inflated obstacles are removed, and a move that the simulator has rejected becomes a forbidden edge.

Magnified search. Small or distant targets are often missed in the full frame. When a view yields no observation with $a_t \geq 0.5$ and nothing is committed, the frame is split

into 3×3 overlapping tiles, each upsampled to 448 px and queried for a box in parallel; up to four distinct tile boxes are crop-checked, and those with $\pi^z \geq 0.5$ enter the memory. This happens once per view and spends VLM calls rather than environment steps.

4.4. Without a depth sensor

Monocular metric depth networks err mostly by a frame-dependent scale. Let $\hat{D}_t$ be the raw prediction and $\kappa(v) = \sin\phi_t + \frac{v-v_0}{\varphi}\cos\phi_t$ the downward component of the ray through row v per unit of planar depth. A pixel on the floor satisfies $\sigma\hat{D}_t(u,v)\,\kappa(v) = h$ for the camera height h, and an object standing on the floor gives a larger σ. Over the pixels $\mathcal{F}_t$ whose rays point at least 15° below the horizontal, S2M takes the mode

$$\sigma_t = \operatorname*{mode}_{(u,v)\in\mathcal{F}_t} \frac{h}{\hat{D}_t(u,v)\,\kappa(v)}, \qquad D_t = \sigma_t\,\hat{D}_t, \tag{11}$$

with the mode taken over values in $[\bar{\sigma}/1.33,\ 1.33\,\bar{\sigma}]$, where $\bar{\sigma}$ is one global scale fitted on development frames; it keeps $\bar{\sigma}$ when fewer than 10% of the pixels in $\mathcal{F}_t$ fall in this window. This is the ground-plane constraint of monocular visual odometry [54] applied per frame. Pose comes from dead reckoning over the executed actions,

$$\hat{x}_{t+1} = \hat{x}_t + 0.25\,o_t\big(\alpha_f e_f(\hat{\theta}_t) + \alpha_r e_r(\hat{\theta}_t)\big), \tag{12}$$

with $(\alpha_f, \alpha_r) \in \{(\pm1,0),(0,\pm1)\}$ the translation of u_t, and heading and tilt advanced by the turn or tilt of u_t when $o_t = 1$.

Explicit stopping. EB-Navigation ends an episode on success; RoboTHOR requires a Stop action. There the agent stops only when it is within 0.9 m of a committed hypothesis, has turned within 20° of it and tilted it into view, and a fresh presence query exceeds 0.5; otherwise the hypothesis is exhausted. Because RoboTHOR episodes last up to 500 steps, negative evidence counts only within 3 m and once per view, and exploration goes to the nearest frontier.

Information access. The VLM sees only the current image, a crop or tile of it, and s; the controller sees depth, pose, and each action's success flag. Neither receives the target position or simulator object state.

5. Experiments

Setup. We evaluate on the full EB-Navigation test set (300 episodes, official environment, actions, and success test) and on the full RoboTHOR ObjectNav validation split [12] (1800 episodes, 15 apartments, 12 categories; stochastic LoCoBot actuation, 30° turns, 500 steps, and a Stop that succeeds within 1 m of a visible target-category object). The models are Qwen3-VL-4B-Instruct [2] and Qwen2.5-VL-3B-Instruct, served with vLLM [29] at temperature 0 on one 8 GB RTX 5060 that also renders the simulator; S2M uses their 4-bit AWQ checkpoints. The EB prompt uses the benchmark's prompts, three examples, parser, and plan execution. Monocular depth comes from Depth Anything V2 Metric-Indoor-Base [67] on the CPU. *Protocol.* The main configuration of S2M was run four times on the test set and the RGB-only configuration twice; a repeated configuration enters every comparison with its per-episode mean. All runs share the same 300 episodes, so all comparisons are paired. We report episode-level bootstrap 95% confidence intervals and two-sided sign-flip permutation tests on the per-episode differences (10^6 samples), Holm-corrected over the ten variants of Tab. 2.

5.1. Main results

With Qwen3-VL-4B, S2M succeeds in 79.4% of the episodes against 55.0% for the EB prompt (Tab. 1), a paired gain of +24.4 pp (CI [+18.0, +30.8], $p<10^{-5}$). The gain is +22.9, +19.2, and +23.3 pp on Base, Common, and Complex and largest where the target starts behind the agent: on Long, S2M succeeds in 82.1% against 25.0%, 1.3 pp below the oracle. On Visual the two agents are level (−0.4 pp, CI [−15.0, +14.6], p=1.00; Sec. 5.4). With Qwen2.5-VL-3B, S2M reaches 70.0% against 20.3% (+49.7 pp, CI [+43.0, +56.3]). Moving from the 4B to the 3B model costs S2M 9.4 pp and the EB prompt 34.7 pp.

Efficiency. On the 140 episodes that the EB prompt and every run of S2M solve, S2M needs 11.1 steps against 12.6 (sign test p=1.9×10^{-4}), and within ten and fifteen steps it has solved 28.2 and 68.5% of the episodes against 16.0 and 44.7% (Fig. 3a). It makes 52.6 short VLM calls per episode against 7.8, with 14.4k prompt and 513 completion tokens against 27.1k and 2.3k.

Comparison with published agents. With a 4B model and no training, S2M exceeds every prompted MLLM up to GPT-4o, the scene-graph, module-graph, skill, memory, harness, and self-evolving agents on GPT-4o and 8–35B models [8, 19, 23, 30, 35, 42, 57], and MuSix and SVA, which train extra models around a 4B policy [21, 65]. Only the tool-library agents [78] and RoboFoundry on GPT-6 Astra score higher (Fig. 5).

Where the gain comes from. Per run, S2M never sees the target in 0.8 episodes (EB prompt: 12), is blocked in 0.5 (26), loses a seen target in 28.5 (43), and ends near it in 9.8 (34) (Fig. 3b, Fig. 4): the look-around, the occupancy map, and the memory of the target remove these failure modes.

5.2. Components

Table 2 changes one element at a time. Disabling multi-view fusion, so that each detection replaces the earlier hypotheses, lowers success by 10.4 pp (CI [−15.0, −6.0], Holm p=8.1×10^{-5}), the only component whose removal survives the correction; the loss falls on the four subsets in which the agent starts facing the target. Removing the look-around

Table 1. Success rate (%) on EB-Navigation test. Top: reported results of prompted MLLMs, training-free agents, and fine-tuned agents. Bottom: our runs on one 8 GB GPU; the EB prompt is the benchmark's agent with the same model; S2M runs the model at 4-bit. ±: standard deviation over repeated test runs. CI: episode-level bootstrap. Bold: best in each column among our runs of the same model.

Method	Model	Base	Common	Complex	Visual	Long	Avg.	95% CI
Prompted MLLMs (EmbodiedBench)								
EB prompt [68]	GPT-4o	55.0	60.0	58.3	60.0	55.0	57.7	–
EB prompt [68]	InternVL3-78B	66.7	63.3	61.7	45.0	31.7	53.7	–
EB prompt [68]	Gemini-2.0-flash	63.3	65.0	50.0	51.7	13.3	48.7	–
EB prompt [68]	Claude-3.7-Sonnet	50.0	61.7	50.0	36.7	26.7	45.0	–
EB prompt [68]	Qwen2.5-VL-72B	46.7	46.7	46.7	26.7	33.3	40.0	–
EB prompt [68]	InternVL3-8B	38.3	30.0	40.0	33.3	5.0	29.3	–
EB prompt [68]	Qwen2.5-VL-7B	20.0	26.7	38.3	16.7	0.0	20.3	–
Training-free agents								
MOSAIC (module graph) [57]	Qwen3-VL-8B	60.0	46.7	45.6	51.1	46.7	50.0	–
ESCA (scene graph) [19]	GPT-4o	66.7	62.0	63.3	55.0	26.7	54.7	–
EmbodiSkill (skills) [23]	Qwen3-VL-32B	65.0	68.0	73.0	67.0	33.0	61.3	–
HODAgent (agent loop) [8]	Qwen3.6-35B-A3B	76.7	55.0	68.3	55.0	36.7	58.3	–
RoboFoundry (self-evolving) [35]	Qwen3.8-27B	76.7	85.0	73.3	63.3	43.3	68.3	–
BrainMem (memory) [42]	GPT-4o	73.3	71.7	80.0	66.6	63.3	71.0	–
EMHO (harness) [30]	Qwen3.8-27B	86.7	76.7	73.3	76.7	63.3	75.3	–
RoboFoundry (self-evolving) [35]	GPT-6 Astra	90.0	85.0	86.7	73.3	66.7	80.3	–
Tool library [78]	Qwen3-VL-8B	85.0	85.0	77.0	75.0	77.0	80.0	–
Tool library [78]	GPT-4o	80.0	85.0	85.0	80.0	83.0	83.0	–
Fine-tuned								
SVA (Q-model) [65]	Qwen3.5-4B	54.2	41.7	45.8	41.7	50.0	46.7	–
MuSix (world models) [21]	Qwen3-VL-4B	63.2	63.2	57.6	62.4	43.2	57.9	–
Ours (one 8 GB GPU)								
EB prompt	Qwen3-VL-4B	66.7	65.0	65.0	**53.3**	25.0	55.0	[49.3, 60.7]
S2M	Qwen3-VL-4B	**89.6**	**84.2**	**88.3**	52.9	**82.1**	**79.4** ±0.5	[75.0, 83.7]
S2M, RGB only†	Qwen3-VL-4B	85.0	80.8	85.8	52.5	78.3	76.5 ±0.2	[71.8, 81.0]
EB prompt	Qwen2.5-VL-3B	21.7	21.7	28.3	20.0	10.0	20.3	[16.0, 25.0]
S2M	Qwen2.5-VL-3B	**78.3**	**71.7**	**71.7**	**53.3**	**75.0**	**70.0**	[64.7, 75.0]
Oracle target‡	–	93.3	93.3	93.3	93.3	83.3	91.3	[88.0, 94.3]

†No depth sensor and no simulator pose (Sec. 5.3). ‡Geometric planner given the true target position: an upper bound.

leaves the average unchanged (0.2 pp) but trades one subset against the others: Long changes by −33.7 pp because the agent no longer turns to find a target behind it, while the other subsets gain the steps that the turn no longer costs (+3.3 to +18.8 pp). Removing exhaustion on arrival changes Visual by −6.2 pp, where appearance-only goals produce look-alikes that must be ruled out once visited. Negative evidence, tiled search, the crop check, and the search index each change success by less than 3 pp, none significantly; they cost no environment steps and target competing candidates, which EB-Navigation's single rooms rarely contain.

5.3. Sensing

Pose. A rejected move leaves the agent in place, and turns and tilts are exact, so dead reckoning (Eq. (12)) reproduces heading and tilt exactly and position up to the simulator's 0.1 m grid snapping: its mean error along the test trajectories is 0.10 m. Accumulated odometry noise of 5% per translation and 2° per turn changes success by −0.1 pp (Tab. 2).

Depth. Monocular depth with one global scale is not enough. With pose from the executed actions and depth from the network under the scale $\bar{\sigma}$, S2M reaches 53.0%, level with the EB prompt (−2.0 pp, p=0.65). The global scale places target detections at a median 0.90 of their true range (Fig. 3c) and lifts 42.5% of the nearby floor above the obstacle threshold, so the agent walks to a point short of the target: 66 failures exhaust a hypothesis within 1 m of it, none with the sensor. The per-frame floor scale (Eq. (11)) lowers the falsely blocked floor to 9.9% and moves the range ratio to 0.93 (sensor: 0.97); success rises to 76.5%, +21.5 pp over the EB prompt ($p<10^{-5}$) and 2.9 pp below the depth sensor (Holm p=0.67).

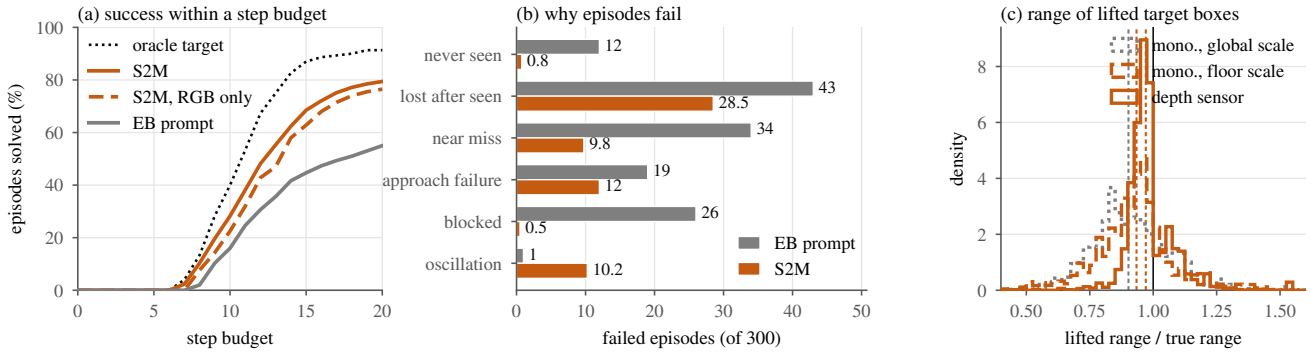


Figure 3. EB-Navigation test, Qwen3-VL-4B. **(a)** Episodes solved within a step budget, averaged over runs. **(b)** Failed episodes by outcome (first matching rule, Sec. 3; mean over the runs of S2M). **(c)** Lifted range divided by true range for detections within 1 m of the target, with medians (dashed), for S2M with the depth sensor and with monocular depth under a global and a per-frame floor scale (Eq. (11)).

Table 2. Components and sensing on EB-Navigation test (Qwen3-VL-4B). Each row changes one element of S2M. Δ: paired difference in average success from the full model, with its 95% CI; p: sign-flip permutation test, Holm-corrected over the ten rows. Bold: significant.

Variant	Base	Common	Complex	Visual	Long	Avg.	Δ (95% CI)		p
S2M (full)	89.6	84.2	88.3	52.9	82.1	79.4	–		–
Components									
w/o fusion	78.3	70.0	75.0	41.7	80.0	69.0	**−10.4**	[−15.0, −6.0]	8.1×10^{-5}
w/o look-around	95.0	91.7	91.7	71.7	48.3	79.7	+0.2	[−4.2, +4.8]	1.00
w/o exhaustion	88.3	85.0	88.3	46.7	83.3	78.3	−1.1	[−2.5, +0.2]	0.89
w/o neg. evidence	93.3	83.3	88.3	58.3	81.7	81.0	+1.6	[−1.2, +4.5]	1.00
w/o tiled search	86.7	80.0	85.0	60.0	80.0	78.3	−1.1	[−4.5, +2.4]	1.00
w/o crop check	85.0	80.0	83.3	51.7	83.3	76.7	−2.8	[−5.7, +0.1]	0.57
strongest first	90.0	85.0	90.0	55.0	83.3	80.7	+1.2	[−0.3, +2.9]	0.89
Sensing									
odometry noise (5%, 2°)	91.7	81.7	90.0	50.0	83.3	79.3	−0.1	[−3.1, +2.9]	1.00
RGB only, global scale	56.7	58.3	60.0	36.7	53.3	53.0	**−26.4**	[−32.4, −20.5]	$<10^{-5}$
RGB only, floor scale	85.0	80.8	85.8	52.5	78.3	76.5	−2.9	[−6.3, +0.3]	0.67

5.4. Remaining gap: appearance-only goals

Visual is the weakest subset in every run of S2M (50.0–56.7%), accounts for 68% of the gap to the oracle, and is the only subset on which S2M does not exceed the EB prompt. The failure is recognition: a candidate comes within 0.5 m of the target in 93–98% of the episodes of every other subset but in 63–67% of the Visual ones, and 65 of the 113 Visual failures never produce one.

5.5. RoboTHOR ObjectNav

On the full RoboTHOR validation split, S2M reaches 59.6% success and 35.9% SPL. Among zero-shot methods (Tab. 3) its SR is above all but ForesightNav, which runs a 12B model [40], and STRIVE, which pairs Gemini-2.0-flash with a dedicated detector [79]; its SPL is second only to STRIVE and above that of the ProcTHOR policy trained for the task [13]. Per category, success ranges from 25.3% (vases) to 88.0% (basketballs; Fig. 6).

6. Conclusion

A small local VLM fails as a navigator when it must carry spatial state through text, and succeeds when it is asked only what the current image can answer. On one 8 GB GPU and without training, it outperforms published agents built on far larger models, also from RGB alone.

Limitations. Without a depth sensor, the scale comes from a visible floor at a known camera height. Appearance-only goals are limited by recognition (Sec. 5.4). Multi-room houses, where the search index matters more, and real robots remain to be tested.

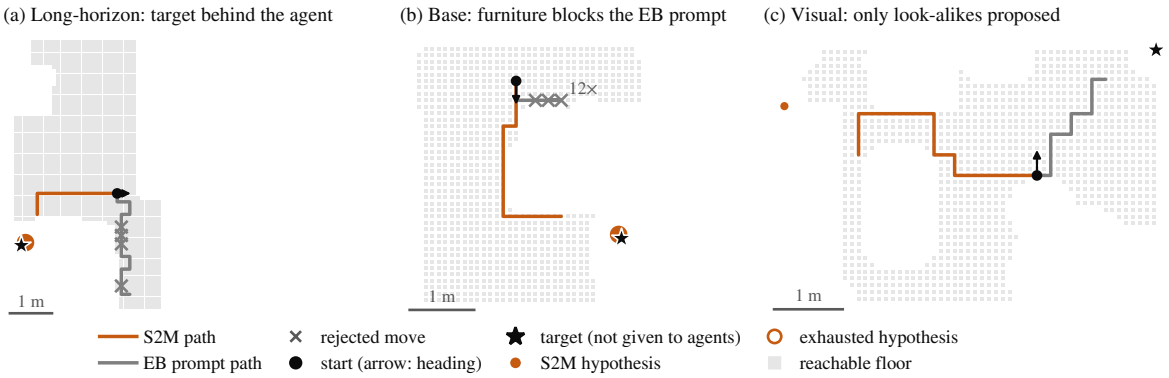


Figure 4. Top-down trajectories of the EB prompt (grey) and S2M (orange) with Qwen3-VL-4B on three test episodes; dots are the hypotheses of S2M at the end of the episode, sized by evidence. (a) The EB prompt never brings the target into view in 20 steps; S2M turns, commits to a hypothesis 0.1 m from the target, and succeeds in 12 steps. (b) 14 of the 20 moves of the EB prompt are rejected; S2M plans around the furniture on its occupancy map and succeeds in 17 steps. (c) A Visual failure of S2M ("Approach the tall green container with a smooth texture."): its hypothesis lies 5.5 m from the target, while the EB prompt succeeds.

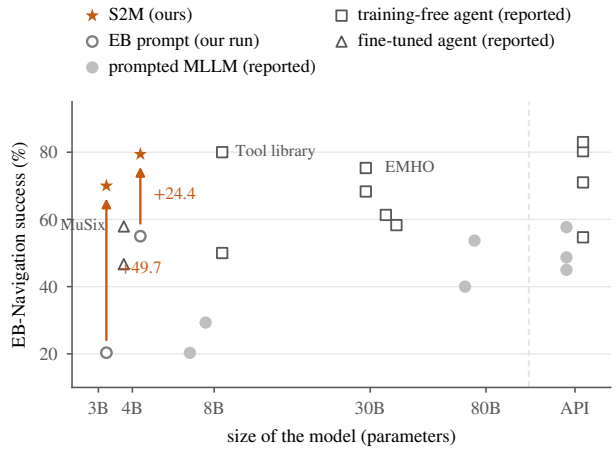


Figure 5. EB-Navigation test success against the size of the model, for every result of Tab. 1. Arrows: S2M against the EB prompt with the same model. Closed models are shown at the right.

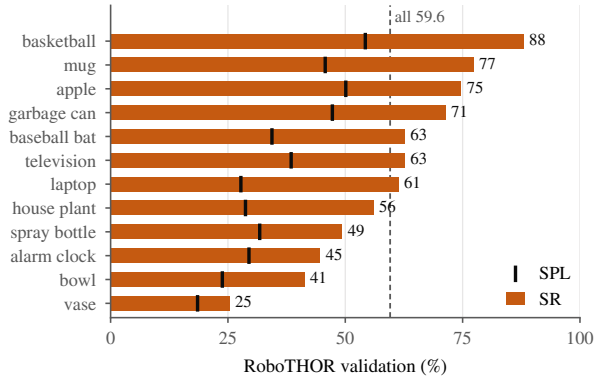


Figure 6. S2M on RoboTHOR validation per target category (bars: SR; ticks: SPL; dashed: all episodes).

Table 3. RoboTHOR ObjectNav validation (1800 episodes). Bold: best; underlined: second best. §As reported by SG-Nav [70].

Method	SR	SPL
Zero-shot		
CoW [15]	26.7	16.9
LGX [14]	35.0	21.9
ESC [77]	38.1	22.2
L3MVN§ [73]	41.2	22.5
VLFM§ [72]	42.3	23.0
OpenFMNav§ [27]	44.1	23.3
SG-Nav [70]	47.5	24.0
UniGoal [71]	48.0	24.2
CogNav [5]	54.6	24.3
MVP-Nav [64]	57.5	26.8
ForesightNav [40]	62.5	32.9
STRIVE [79]	**68.1**	**36.3**
Trained for the task		
ProcTHOR [13, 15]	<u>66.4</u>	27.4
S2M (Qwen3-VL-4B)	59.6	<u>35.9</u>

References

[1] Abrar Anwar, John Welsh, Joydeep Biswas, Soha Pouya, and Yan Chang. ReMEmbR: Building and Reasoning Over Long-Horizon Spatio-Temporal Memory for Robot Navigation, 2024. ICRA 2025, arXiv:2409.13682. 3

[2] Shuai Bai, Yuxuan Cai, Ruizhe Chen, Keqin Chen, Xionghui Chen, et al. Qwen3-VL Technical Report, 2025. arXiv preprint, arXiv:2511.21631. 5

[3] Shuai Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, et al. Qwen2.5-VL Technical Report, 2025. arXiv preprint, arXiv:2502.13923. 1

[4] Wenzhe Cai, Jiaqi Peng, Yuqiang Yang, Yujian Zhang, Meng Wei, Hanqing Wang, Yilun Chen, Tai Wang, and Jiangmiao Pang. NavDP: Learning Sim-to-Real Navigation Diffusion Policy with Privileged Information Guidance, 2025. ICRA 2026, arXiv:2505.08712. 3

[5] Yihan Cao, Jiazhao Zhang, Zhinan Yu, Shuzhen Liu, Zheng Qin, Qin Zou, Bo Du, and Kai Xu. CogNav: Cognitive Process Modeling for Object Goal Navigation with LLMs, 2024. ICCV 2025, arXiv:2412.10439. 3, 8

[6] Hanyang Chen, Mark Zhao, Rui Yang, Qinwei Ma, Ke Yang, Jiarui Yao, Kangrui Wang, Hao Bai, Zhenhailong Wang, Rui Pan, Mengchao Zhang, Jose Barreiros, Aykut Onol, ChengXiang Zhai, Heng Ji, Manling Li, Huan Zhang, and Tong Zhang. ERA: Transforming VLMs into Embodied Agents via Embodied Prior Learning and Online Reinforcement Learning, 2025. arXiv Oct 2025, arXiv:2510.12693. 2

[7] Minghao Chen, Yihang Li, Yanting Yang, Shiyu Yu, Binbin Lin, and Xiaofei He. AutoManual: Constructing Instruction Manuals by LLM Agents via Interactive Environmental Learning, 2024. NeurIPS 2024, arXiv:2405.16247. 2

[8] Wang Warren Chen, Jiahao Zhang, Zhenjiang Li, Mingxu Wang, Lei Yi, Yuchen Kang, Shuo Sun, Ziping Chen, and Jie Chen. HODAgent: Towards on-demand, responsive humanoids for physical world human interaction, 2026. arXiv:2608.17584. 5, 6

[9] Hyeonbeom Choi, Daechul Ahn, Youhan Lee, Taewook Kang, Seongwon Cho, and Jonghyun Choi. SCALE: Self-uncertainty Conditioned Adaptive Looking and Execution for Vision-Language-Action Models, 2026. ICML 2026, arXiv:2602.04208. 3

[10] Roberto Corlito, Fabian Schmidt, Nils Seibert, Markus Enzweiler, Abhinav Valada, and Arne Roennau. Sensor Configuration Matters: A Systematic Evaluation of Multimodal SLAM on Quadruped Robots, 2026. arXiv 2026, arXiv:2606.19067. 3

[11] Jadelynn Dao, Milan Ganai, Yasmina Abukhadra, Ajay Sridhar, Mozhgan Nasr Azadani, Katie Luo, Clark Barrett, Jiajun Wu, Chelsea Finn, and Marco Pavone. DIRECT: When and Where Should You Allocate Test-Time Compute in Embodied Planners?, 2026. arXiv Jun 2026, arXiv:2606.12402. 3

[12] Matt Deitke, Winson Han, Alvaro Herrasti, Aniruddha Kembhavi, Eric Kolve, Roozbeh Mottaghi, Jordi Salvador, Dustin Schwenk, Eli VanderBilt, Matthew Wallingford, Luca Weihs, Mark Yatskar, and Ali Farhadi. RoboTHOR: An Open Simulation-to-Real Embodied AI Platform, 2020. CVPR 2020, arXiv:2004.06799. 1, 5

[13] Matt Deitke, Eli VanderBilt, Alvaro Herrasti, Luca Weihs, Jordi Salvador, Kiana Ehsani, Winson Han, Eric Kolve, Ali Farhadi, Aniruddha Kembhavi, and Roozbeh Mottaghi. ProcTHOR: Large-scale embodied AI using procedural generation. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022. 7, 8

[14] Vishnu Sashank Dorbala, James F. Mullen, and Dinesh Manocha. Can an embodied agent find your "cat-shaped mug"? LLM-guided exploration for zero-shot object navigation. *IEEE Robotics and Automation Letters*, 9(5):4083–4090, 2024. 8

[15] Samir Yitzhak Gadre, Mitchell Wortsman, Gabriel Ilharco, Ludwig Schmidt, and Shuran Song. CoWs on Pasture: Baselines and Benchmarks for Language-Driven Zero-Shot Object Navigation, 2022. CVPR 2023, arXiv:2203.10421. 2, 8

[16] Muhammad Fadhil Ginting, Dong-Ki Kim, Xiangyun Meng, Andrzej Reinke, Bandi Jai Krishna, Navid Kayhani, Oriana Peltzer, David D. Fan, Amirreza Shaban, Sung-Kyun Kim, Mykel J. Kochenderfer, Ali akbar Agha-mohammadi, and Shayegan Omidshafiei. Enter the Mind Palace: Reasoning and Planning for Long-term Active Embodied Question Answering, 2025. arXiv Jul 2025, v2 Sep 2025, arXiv:2507.12846. 3

[17] Dylan Goetting, Himanshu Gaurav Singh, and Antonio Loquercio. End-to-End Navigation with Vision Language Models: Transforming Spatial Reasoning into Question-Answering, 2024. arXiv 2411.05755, 2024, arXiv:2411.05755. 3

[18] Qiao Gu, Alihusein Kuwajerwala, Sacha Morin, Krishna Murthy Jatavallabhula, Bipasha Sen, Aditya Agarwal, Corban Rivera, William Paul, Kirsty Ellis, Rama Chellappa, Chuang Gan, Celso Miguel de Melo, Joshua B. Tenenbaum, Antonio Torralba, Florian Shkurti, and Liam Paull. ConceptGraphs: Open-Vocabulary 3D Scene Graphs for Perception and Planning, 2023. ICRA 2024, arXiv:2309.16650. 3

[19] Jiani Huang, Amish Sethi, Matthew Kuo, Mayank Keoliya, Neelay Velingker, JungHo Jung, Ser-Nam Lim, Ziyang Li, and Mayur Naik. ESCA: Contextualizing embodied agents via scene-graph generation. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025. 5, 6

[20] Xun Huang, Shijia Zhao, Yunxiang Wang, Xin Lu, Wanfa Zhang, Rongsheng Qu, Weixin Li, Yunhong Wang, and Chenglu Wen. MSGNav: Unleashing the Power of Multi-modal 3D Scene Graph for Zero-Shot Embodied Navigation, 2025. CVPR 2026, arXiv:2511.10376. 2

[21] Jinwoo Jang, Daniel J. Rho, Sihyung Yoon, Hyunsuk Cho, and Honguk Woo. Multi-scale mixture of world models for embodied agents in evolving environments, 2026. arXiv:2607.00457. 5, 6

[22] Jiajun Jiang, Yiming Zhu, Zirui Wu, and Jie Song. DualMap: Online Open-Vocabulary Semantic Mapping for Natural Language Navigation in Dynamic Changing Scenes, 2025. IEEE RA-L 2025, arXiv:2506.01950. 3

[23] Ruofei Ju, Xinrui Wang, Xin Ding, Yifan Yang, Hao Wu, Shiqi Jiang, Qianxi Zhang, Hao Wen, Xiangyu Li, Weijun Wang, Kun Li, Yunxin Liu, Haipeng Dai, Wei Wang, and Ting Cao. EmbodiSkill: Skill-aware reflection for self-evolving embodied agents, 2026. arXiv:2605.10332. 5, 6

[24] Joseph B. Kadane and Herbert A. Simon. Optimal Strategies for a Class of Constrained Sequential Problems. *The Annals of Statistics*, 5(2):237–255, 1977. 3, 4

[25] Saimouli Katragadda, Cho-Ying Wu, Yuliang Guo, Xinyu Huang, Guoquan Huang, and Liu Ren. Online Language Splatting, 2025. arXiv 2025, arXiv:2503.09447. 3

[26] Eric Kolve, Roozbeh Mottaghi, Winson Han, Eli VanderBilt, Luca Weihs, Alvaro Herrasti, Matt Deitke, Kiana Ehsani, Daniel Gordon, Yuke Zhu, Aniruddha Kembhavi, Abhinav Gupta, and Ali Farhadi. AI2-THOR: An Interactive 3D Environment for Visual AI, 2017. arXiv preprint, arXiv:1712.05474. 3

[27] Yuxuan Kuang, Hai Lin, and Meng Jiang. OpenFMNav: Towards open-set zero-shot object navigation via vision-language foundation models. In *Findings of the Association for Computational Linguistics: NAACL*, 2024. 8

[28] Jacky Kwok, Christopher Agia, Rohan Sinha, Matt Foutter, Shulu Li, Ion Stoica, Azalia Mirhoseini, and Marco Pavone. RoboMonkey: Scaling Test-Time Sampling and Verification for Vision-Language-Action Models, 2025. arXiv 2506.17811, 2025, arXiv:2506.17811. 3

[29] Woosuk Kwon, Zhuohan Li, Siyuan Zhuang, Ying Sheng, Lianmin Zheng, Cody Hao Yu, Joseph E. Gonzalez, Hao Zhang, and Ion Stoica. Efficient Memory Management for Large Language Model Serving with PagedAttention, 2023. SOSP 2023, arXiv:2309.06180. 5

[30] Hyun Jung Lee, Jungtaek Kim, Jongwon Jeong, Tae-Eui Kam, Donghyun Kim, and Yong Jae Lee. EMHO: EMbodied Agent Harness Optimization via Experience Traces, 2026. arXiv 2610.08432, Oct 2026, arXiv:2610.08432. 1, 2, 5, 6

[31] Seungjun Lee, Zihan Wang, Yunsong Wang, and Gim Hee Lee. EmbodiedSplat: Online Feed-Forward Semantic 3DGS for Open-Vocabulary 3D Scene Understanding, 2026. CVPR 2026, arXiv:2603.04254. 3

[32] Mingcong Lei, Honghao Cai, Yuyuan Yang, Yimou Wu, Jinke Ren, Zezhou Cui, Liangchen Tan, Junkun Hong, Gehan Hu, Shuangyu Zhu, Shaohan Jiang, Ge Wang, Junyuan Tan, Zhenglin Wan, Zheng Li, Zhen Li, Shuguang Cui, Yiming Zhao, and Yatong Han. RoboMemory: A Brain-inspired Multi-memory Agentic Framework for Interactive Environmental Learning in Physical Embodied Systems, 2025. arXiv 2508.01415, 2025, arXiv:2508.01415. 2

[33] Manling Li, Shiyu Zhao, Qineng Wang, Kangrui Wang, Yu Zhou, Sanjana Srivastava, Cem Gokmen, Tony Lee, Li Erran Li, Ruohan Zhang, Weiyu Liu, Percy Liang, Li Fei-Fei, Jiayuan Mao, and Jiajun Wu. Embodied Agent Interface: Benchmarking LLMs for Embodied Decision Making, 2024. NeurIPS 2024 Datasets & Benchmarks, arXiv:2410.07166. 2

[34] Zeqi Li, Xunyi Zhao, Zerui Li, Sihao Lin, Qi Wu, and ShiJie Li. Structure over Reasoning: Unified Zero-Shot Navigation with a Small Vision-Language Model, 2026. arXiv preprint, arXiv:2610.25130. 3

[35] Jingsong Liang, Shuhao Liao, Shizhe Zhang, Diyuan Hou, Yuxin Cai, Xinjian Deng, Chengyang He, Wenhui Huang, Runjia Tan, Zhidong Wang, Lan Yu, Xuesong Tian, Guillaume Sartoretti, Jie Luo, Yao Mu, Wenjun Wu, Wanhua Li, and Chen Lv. RoboFoundry: System-as-policy evolution for self-learning embodied agents, 2026. arXiv:2609.32862. 5, 6

[36] Kaiqu Liang, Zixu Zhang, and Jaime Fernández Fisac. Introspective Planning: Aligning Robots' Uncertainty with Inherent Task Ambiguity, 2024. NeurIPS 2024, arXiv:2402.06529. 3

[37] Haotong Lin, Sili Chen, Junhao Liew, Donny Y. Chen, Zhenyu Li, Guang Shi, Jiashi Feng, and Bingyi Kang. Depth Anything 3: Recovering the Visual Space from Any Views, 2025. arXiv 2025, arXiv:2511.10647. 3

[38] Guanlin Liu, Shaobin Ling, Renyuan Liu, Zeying Gong, and Junjie Hu. AECNav: Active Evidence Consolidation for Efficient Zero-Shot Open-Vocabulary Object Navigation, 2026. arXiv preprint, arXiv:2608.10817. 3

[39] Jinrui Liu, Bingyan Nie, Boyu Li, Yaran Chen, Yuze Wang, Shunsen He, and Haoran Li. RoboGPT-R1: Enhancing Robot Task Planning with Reinforcement Learning, 2025. AAMAS 2026, arXiv:2510.14828. 2

[40] Yu Liu, Chengjie Fang, and Likun Lu. ForesightNav: Adaptive semantic-geometric planning for universal zero-shot goal-oriented navigation. *IEEE Access*, 14, 2026. 7, 8

[41] Yuxing Long, Wenzhe Cai, Hongcheng Wang, Guanqi Zhan, and Hao Dong. InstructNav: Zero-shot System for Generic Instruction Navigation in Unexplored Environment, 2024. arXiv 2406.04882, arXiv:2406.04882. 3

[42] Xiaoyu Ma, Lianyu Hu, Wenbing Tang, Zixuan Hu, Zeqin Liao, Zhizhen Wu, and Yang Liu. BrainMem: Brain-Inspired Evolving Memory for Embodied Agent Task Planning, 2026. arXiv 2604.16331, Mar 2026, arXiv:2604.16331. 1, 2, 5, 6

[43] Dominic Maggio, Hyungtae Lim, and Luca Carlone. VGGT-SLAM: Dense RGB SLAM Optimized on the SL(4) Manifold, 2025. NeurIPS 2025, arXiv:2505.12549. 3

[44] Riku Murai, Eric Dexheimer, and Andrew J. Davison.

MASt3R-SLAM: Real-Time Dense SLAM with 3D Reconstruction Priors, 2024. CVPR 2025, arXiv:2412.12392. 3

[45] Soroush Nasiriany, Fei Xia, Wenhao Yu, Ted Xiao, Jacky Liang, Ishita Dasgupta, Annie Xie, Danny Driess, Ayzaan Wahid, Zhuo Xu, Quan Vuong, Tingnan Zhang, Tsang-Wei Edward Lee, Kuang-Huei Lee, Peng Xu, Sean Kirmani, Yuke Zhu, Andy Zeng, Karol Hausman, Nicolas Heess, Chelsea Finn, Sergey Levine, and Brian Ichter. PIVOT: Iterative Visual Prompting Elicits Actionable Knowledge for VLMs, 2024. arXiv 2402.07872, 2024, arXiv:2402.07872. 3

[46] Dujun Nie, Xianda Guo, Yiqun Duan, Ruijun Zhang, and Long Chen. WMNav: Integrating Vision-Language Models into World Models for Object Goal Navigation, 2025. IROS 2025 Oral, arXiv:2503.02247. 3

[47] Yanyuan Qiao, Haodong Hong, Wenqi Lyu, Dong An, Siqi Zhang, Yutong Xie, Xinyu Wang, and Qi Wu. NavBench: Probing Multimodal Large Language Models for Embodied Navigation, 2025. arXiv 2506.01031, 2025, arXiv:2506.01031. 2

[48] Minghan Qin, Wanhua Li, Jiawei Zhou, Haoqian Wang, and Hanspeter Pfister. LangSplat: 3D Language Gaussian Splatting, 2023. CVPR 2024, arXiv:2312.16084. 3

[49] Krishan Rana, Jesse Haviland, Sourav Garg, Jad Abou-Chakra, Ian Reid, and Niko Suenderhauf. SayPlan: Grounding Large Language Models using 3D Scene Graphs for Scalable Robot Task Planning, 2023. CoRL 2023, arXiv:2307.06135. 3

[50] Allen Z. Ren, Anushri Dixit, Alexandra Bodrova, Sumeet Singh, Stephen Tu, Noah Brown, Peng Xu, Leila Takayama, Fei Xia, Jake Varley, Zhenjia Xu, Dorsa Sadigh, Andy Zeng, and Anirudha Majumdar. Robots That Ask For Help: Uncertainty Alignment for Large Language Model Planners, 2023. CoRL 2023, oral, arXiv:2307.01928. 3

[51] Allen Z. Ren, Jaden Clark, Anushri Dixit, Masha Itkina, Anirudha Majumdar, and Dorsa Sadigh. Explore until Confident: Efficient Exploration for Embodied Question Answering, 2024. RSS 2024, arXiv:2403.15941. 3

[52] Noah Shinn, Federico Cassano, Edward Berman, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. Reflexion: Language Agents with Verbal Reinforcement Learning, 2023. NeurIPS 2023, arXiv:2303.11366. 2

[53] Nishad Singhi, Christian Bialas, Snehal Jauhri, Vignesh Prasad, Georgia Chalvatzaki, Marcus Rohrbach, and Anna Rohrbach. Think Twice, Act Once: Verifier-Guided Action Selection For Embodied Agents, 2026. CVPR 2026 Findings, arXiv:2605.12620. 2

[54] Shiyu Song and Manmohan Chandraker. Robust Scale Estimation in Real-Time Monocular SFM for Autonomous Driving. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2014. 3, 5

[55] Jianyuan Wang, Minghao Chen, Nikita Karaev, Andrea Vedaldi, Christian Rupprecht, and David Novotny. VGGT: Visual Geometry Grounded Transformer, 2025. CVPR 2025, arXiv:2503.11651. 3

[56] Kangrui Wang, Pingyue Zhang, Zihan Wang, Yaning Gao, Linjie Li, Qineng Wang, Hanyang Chen, Chi Wan, Yiping Lu, Zhengyuan Yang, Lijuan Wang, Ranjay Krishna, Jiajun Wu, Li Fei-Fei, Yejin Choi, and Manling Li. VAGEN: Reinforcing World Model Reasoning for Multi-Turn VLM Agents, 2025. NeurIPS 2025, arXiv:2510.16907. 2

[57] Kevin Wang et al. MOSAIC: The right modules for each task in embodied agents, 2026. OpenReview, https://openreview.net/forum?id=pF9wISLUYo. 5, 6

[58] Liuyi Wang, Xinyuan Xia, Hui Zhao, Hanqing Wang, Tai Wang, Yilun Chen, Chengju Liu, Qijun Chen, and Jiangmiao Pang. Rethinking the Embodied Gap in Vision-and-Language Navigation: A Holistic Study of Physical and Visual Disparities, 2025. ICCV 2025, arXiv:2507.13019. 3

[59] Zhengcheng Wang, Zichuan Lin, Yijun Yang, Haobo Fu, and Deheng Ye. SeeNav-Agent: Enhancing Vision-Language Navigation with Visual Prompt and Step-Level Policy Optimization, 2025. arXiv 2512.02631, Dec 2025, arXiv:2512.02631. 2

[60] Zora Zhiruo Wang, Jiayuan Mao, Daniel Fried, and Graham Neubig. Agent Workflow Memory, 2024. arXiv 2409.07429, 2024, arXiv:2409.07429. 2

[61] Bowen Wen, Matthew Trepte, Joseph Aribido, Jan Kautz, Orazio Gallo, and Stan Birchfield. FoundationStereo: Zero-Shot Stereo Matching, 2025. CVPR 2025, arXiv:2501.09898. 3

[62] Abdelrhman Werby, Chenguang Huang, Martin Büchner, Abhinav Valada, and Wolfram Burgard. Hierarchical Open-Vocabulary 3D Scene Graphs for Language-Grounded Robot Navigation, 2024. RSS 2024, arXiv:2403.17846. 3

[63] Abdelrhman Werby, Chenguang Huang, Martin Büchner, Abhinav Valada, and Wolfram Burgard. Hierarchical Open-Vocabulary 3D Scene Graphs for Language-Grounded Robot Navigation, 2024. RSS 2024, arXiv:2403.17846. 3

[64] Wenyuan Xie, Shaokai Wu, Yijin Zhou, Yanbiao Ji, Guodong Zhang, Bayram Bayramli, Qiuchang Li, Xunchu Zhou, Yue Ding, and Hongtao Lu. MVP-Nav: Multi-layer Value Map Planner Navigator, 2026. arXiv:2606.31919. 8

[65] Xinyi Xie, Zican Hu, Zhanyu Liu, Yicheng Dong, Wenhao Wu, Zhenhong Sun, Haoran Li, Chunlin Chen, Zhi Wang, and Pichao Wang. Look before you leap: Distilling tree search into action evaluation for frozen VLA models, 2026. arXiv:2607.03751. 5, 6

[66] Jianwei Yang, Hao Zhang, Feng Li, Xueyan Zou, Chunyuan Li, and Jianfeng Gao. Set-of-Mark Prompting Unleashes Extraordinary Visual Grounding in GPT-4V, 2023. arXiv 2310.11441, 2023, arXiv:2310.11441. 3

[67] Lihe Yang, Bingyi Kang, Zilong Huang, Zhen Zhao, Xiaogang Xu, Jiashi Feng, and Hengshuang Zhao. Depth Anything V2, 2024. NeurIPS 2024, arXiv:2406.09414. 3, 5

[68] Rui Yang, Hanyang Chen, Junyu Zhang, Mark Zhao, Cheng Qian, Kangrui Wang, Qineng Wang, Teja Venkat Koripella, Marziyeh Movahedi, Manling Li, Heng Ji, Huan Zhang, and Tong Zhang. EmbodiedBench: Comprehensive Benchmarking Multi-modal Large Language Models for Vision-Driven Embodied Agents, 2025. ICML 2025, arXiv:2502.09560. 1, 2, 3, 6

[69] Yuncong Yang, Han Yang, Jiachen Zhou, Peihao Chen, Hongxin Zhang, Yilun Du, and Chuang Gan. 3D-Mem: 3D Scene Memory for Embodied Exploration and Reasoning, 2024. CVPR 2025, arXiv:2411.17735. 3

[70] Hang Yin, Xiuwei Xu, Zhenyu Wu, Jie Zhou, and Jiwen Lu. SG-Nav: Online 3D Scene Graph Prompting for LLM-based Zero-shot Object Navigation, 2024. NeurIPS 2024, arXiv:2410.08189. 3, 8

[71] Hang Yin, Xiuwei Xu, Lingqing Zhao, Ziwei Wang, Jie Zhou, and Jiwen Lu. UniGoal: Towards Universal Zero-shot Goal-oriented Navigation, 2025. CVPR 2025, arXiv:2503.10630. 3, 8

[72] Naoki Yokoyama, Sehoon Ha, Dhruv Batra, Jiuguang Wang, and Bernadette Bucher. VLFM: Vision-Language Frontier Maps for Zero-Shot Semantic Navigation, 2023. ICRA 2024, arXiv:2312.03275. 2, 8

[73] Bangguo Yu, Hamidreza Kasaei, and Ming Cao. L3MVN: Leveraging large language models for visual target navigation. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2023. 8

[74] Mingjie Zhang, Yuheng Du, Chengkai Wu, Jinni Zhou, Zhenchao Qi, Jun Ma, and Boyu Zhou. ApexNav: An Adaptive Exploration Strategy for Zero-Shot Object Navigation with Target-centric Semantic Fusion, 2025. IEEE RA-L 2025, arXiv:2504.14478. 3

[75] Wenqi Zhang, Mengna Wang, Gangao Liu, Xu Huixin, Yiwei Jiang, Yongliang Shen, Guiyang Hou, Zhe Zheng, Hang Zhang, Xin Li, Weiming Lu, Peng Li, and Yueting Zhuang. Embodied-Reasoner: Synergizing Visual Search, Reasoning, and Action for Embodied Interactive Tasks, 2025. arXiv 2503.21696, 2025, arXiv:2503.21696. 2

[76] Andrew Zhao, Daniel Huang, Quentin Xu, Matthieu Lin, Yong-Jin Liu, and Gao Huang. ExpeL: LLM Agents Are Experiential Learners, 2023. AAAI 2024, arXiv:2308.10144. 2

[77] Kaiwen Zhou, Kaizhi Zheng, Connor Pryor, Yilin Shen, Hongxia Jin, Lise Getoor, and Xin Eric Wang. ESC: Exploration with soft commonsense constraints for zero-shot object navigation. In *International Conference on Machine Learning (ICML)*, 2023. 8

[78] Xueyang Zhou, Zijia Wang, Qianjiang Li, Yibo Hu, Guiyao Tie, Li Wan, Yidan Liu, Pan Zhou, Lichao Sun, and Yongchao Chen. Enabling Extensible Embodied Capabilities with Tools, 2026. arXiv 2605.26637, May 2026, arXiv:2605.26637. 1, 2, 5, 6

[79] Haokun Zhu, Zongtai Li, Zhixuan Liu, Wenshan Wang, Ji Zhang, Jonathan Francis, and Jean Oh. STRIVE: Structured Representation Integrating VLM Reasoning for Efficient Object Navigation, 2025. arXiv preprint, arXiv:2505.06729. 3, 7, 8